

Education

KTH Royal Institute of Technology

MSc, Interactive Media Technology

Thesis: *Redesigning No-Code Virtual Reality Platform for K12 Education with Teachers' Tacit Knowledge*

Sweden

Ongoing

École Polytechnique Fédérale de Lausanne (EPFL)

Graduate Study Abroad, School of Computer and Communication Sciences

Switzerland

2021 – 2023

West Bengal University of Technology

BTech, Information Technology

India

2019

Indian Institute of Technology Kharagpur

Undergraduate Visiting Student, Department of Computer Science and Engineering

Thesis: *Counterfeit Integrated Circuit Detection using Computer Vision and Image Processing Techniques*

India

2017 – 2018

Mentoring Experience

Teaching Assistant (UG & PG) • KTH Royal Institute of Technology

Sep 2020 – Jun 2025

- Machine Learning • Programming Fundamentals • Multivariable Calculus • Computer Networking • Visualisation

Management Consulting Case Mentor (UG & PG) • EPFL

Aug 2022 – Jan 2023

- EPFL Consulting Society: Mentored students of EPFL, UNIL, HEC Lausanne of diverse backgrounds in weekly studies for 1 semester

Visiting Lecturer (UG) • NSHM Knowledge Campus, Kolkata

Aug 2019 – Dec 2019

- Interactive 3D Computer Graphics using OpenGL and other Languages (WBUT AICTE and Non-AICTE syllabus followed)

Experience in UX Research

Accessible Graph Navigation for Blind Users

Jyväskylän Yliopisto

Finland

Aug 2025

- Identified key accessibility barriers in real-time independent graph navigation, interviewing blind users who took GRE in past
- Following Eisenhower Decision Principles & a Kanban board, prioritized graph sonification (interview) over haptic-thermal features (soma design) to maximize real-time accessibility impact within a 7-day timeline, hardware, & user access constraints
- Prototyped graph sonification using Python to enable real-time interactive graph navigation following heuristic UX principles

Assistive Technology for Hybrid Programming Classes

University of Lausanne

Switzerland

Dec 2022 – Feb 2023

- Facilitated pain-point discovery through semi-structured interviews with 38 students, translating hybrid group-workflow insights into prioritized student-centric product opportunities for Jupyter Thebe, programming assistant for Jupyter Notebook
- Drove iterative product discovery through a participatory design workshop with 32 students (8 hybrid groups of 4 students who used Thebe in former programming classes) with assistance of a doctoral researcher, resulting in 24 designs of Thebe UI
- Synthesized UI findings into 10 design principles to develop assistive technologies for hybrid programming learning activities

Impact of Multimodal Workplace Technologies on Independent Thinking

Kungliga Tekniska Högskolan

Sweden

Dec 2020 – Feb 2021

- Conducted task-based semi-structured interviews to examine impact of used workplace technology (Zoom, Oculus VTime XR, Telepresence Robot Furhat controlled by Oculus HMD) on independent thinking during online and hybrid group collaborations
- Prototyped telepresence robot using Unity3D, C# & Python to manipulate Furhat's head, gaze & voice remotely using Oculus
- Co-designed a survey to understand participants' perceptions of the used telecommunication media on intention to use them
- Analyzed multimodal perceptions & thinking ability during the interactions using mixed-methods analysis on collected data

Experience in Immersive Technology

Emotion Recognition from Facial Expression in Virtual Reality using Machine Learning

EPFL

Switzerland

Sep 2021 – Feb 2022

- Co-developed a real-time emotion-recognition pipeline in VR, reaching 45% accuracy in LightGBM and 41% in Random Forest
- Developed a Unity3D-based prototype following FACS (Facial Action Coding System) for avatar facial expression data collection
- Collected data of 27 subjects & developed a database of avatar blendshapes & raw lips using HTC Vibe HMD & facial tracker

Haptic Perception in Teleoperation Robotic Systems

Sweden

Ericsson

Jun 2021 – Sep 2021

- Co-developed a haptic teleoperation system assisting 1 colleague, connecting 2 Omni haptic pens and Ericsson internal server
- Led experiment design and front-end development for a delivered prototype that uses a modular LEGO object to manipulate physical load, eliciting the impact of network variables (latency, jitter, camera constraints) on haptic perception via A/B testing

Rogue-like Video Game Mike

Sweden

KTH

Jun 2021 – Sep 2021

- Directed Scrum-based development using Unreal Engine and C# with 4 teammates to ship the rogue-like game prototype
- Led task allocation & sprint planning, conducting competitive market research \& heuristic evaluation for design decisions

Other Relevant Experience

Emotion Perception from Gaze

Switzerland

EPFL

Sep 2021 – Feb 2022

- Pre-trained model on ETH-XGaze & transferred it to RAF-DB data to generate pseudo-gaze labels across 7 emotion categories
- Built a gaze estimation pipeline using ResNet-18 for spatial feature extraction combined with a 6-layer Transformer Encoder
- Engineered a preprocessing pipeline using Python based on eye-to-nose landmark ratios to isolate face-forward subjects and to calculate projection thresholds to classify direct vs. averted gazes, for mapping them to approach vs. avoidance emotions
- Integrated a learnable positional token within transformer architecture to map output states to final linear regression layer

Designing Multiuser Phygital Civic Engagement Platform

Switzerland

EPFL

Sep 2021 – Feb 2022

- Led product strategy as part of a group of 5 members (2 Architects, 3 computer engineers) for scalable phygital civic platform
- Designed an AI recommendation engine with API-integrated feedback loops that uses semantic text analysis of user activities
- Strategized end-to-end system workflows for backend API, System Design & UI view for every end-user using Sketch and Miro

Skills

- **UX Research:** Diary Study, A/B Testing, Soma Design, Contextual Inquiry, Survey, Interviewing, Design Fiction, Co-design Workshop
- **Programming & Tools:** Unity3D, Python, C/C++, PyTorch, TensorFlow, OpenCV, GLSL, Plotly, Jax, Git, Jira, Notion, Figma, Miro